

## MULTI-SEGMENTAL EXTENSION



### Test Objective for the Multi-Segmental Extension:

The Multi-Segmental Extension assessment tests for normal extension of hips and spine while maintaining full shoulder flexion.

### How to Perform the Multi-Segmental Extension:

Have the patient assume the starting position by standing erect with feet together and toes pointing forward. The patient then raises their hands into full shoulder flexion, with the elbows extended and in line with the ears. Have the patient push their hips forward and bend backward as far as possible, making sure the hips go forward and the arms go back simultaneously. Direct the patient to look up towards hands during the movement.

Special Note: If the arms do not start in the proper start position, the test is already dysfunctional. Proceed through the motion to only check for pain.

### Functional Criteria:

- Shoulder reach and maintain 170 degrees of shoulder flexion
- ASIS clears the toes
- Spine of scapula clears the heels
- Uniform spine curve
- Without excessive effort and/or motor control

### Common Compensations:

- Greater than 5 degrees of knee Flexion

### Tips for Testing:

- Observe from the front and side.
- Foot position should remain unchanged throughout the movement.
- Do not coach the movement; simply repeat the instructions if needed.
- Was there pain?



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Selective Breakout Tests for Multi-Segmental Extension:

#### Spine/Upper Body



Prone Press Up



Active Lumbar  
Locked (IR) Extension/  
Rotation Test



Passive Lumbar  
Locked (IR) Extension/  
Rotation Test



Active Prone on  
Elbow Extension/  
Rotation Test



Passive Prone on  
Elbow Extension/  
Rotation Test



Active Prone  
Shoulder  
Girdle Flexion  
Test



Passive Prone  
Shoulder Girdle  
Flexion Test

#### Hips



FABER Test



Stabilized FABER  
Test



Modified  
Thomas Test



Active Prone  
Hip Extension

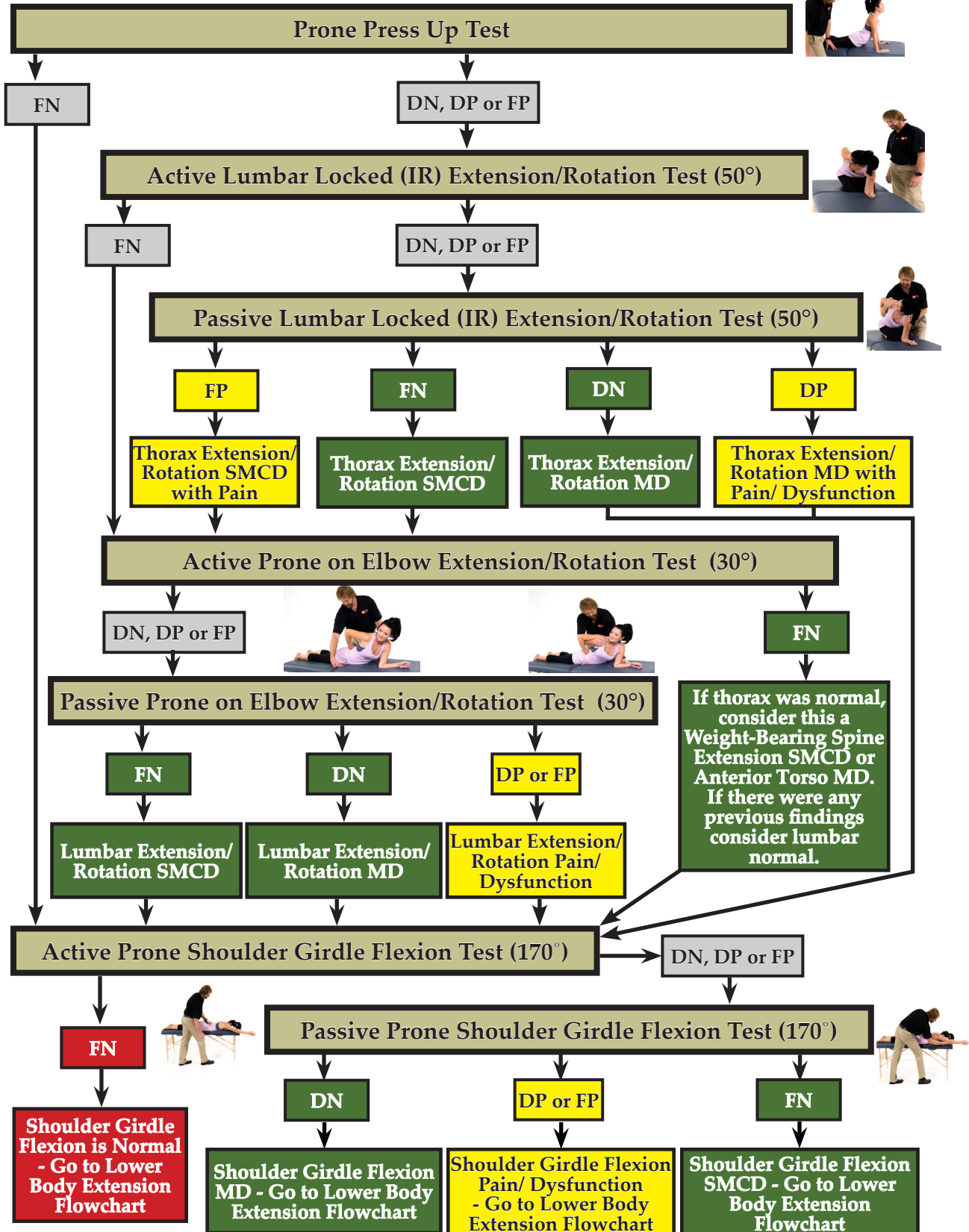


Passive Prone  
Hip Extension

# MULTI-SEGMENTAL EXTENSION BREAKOUT



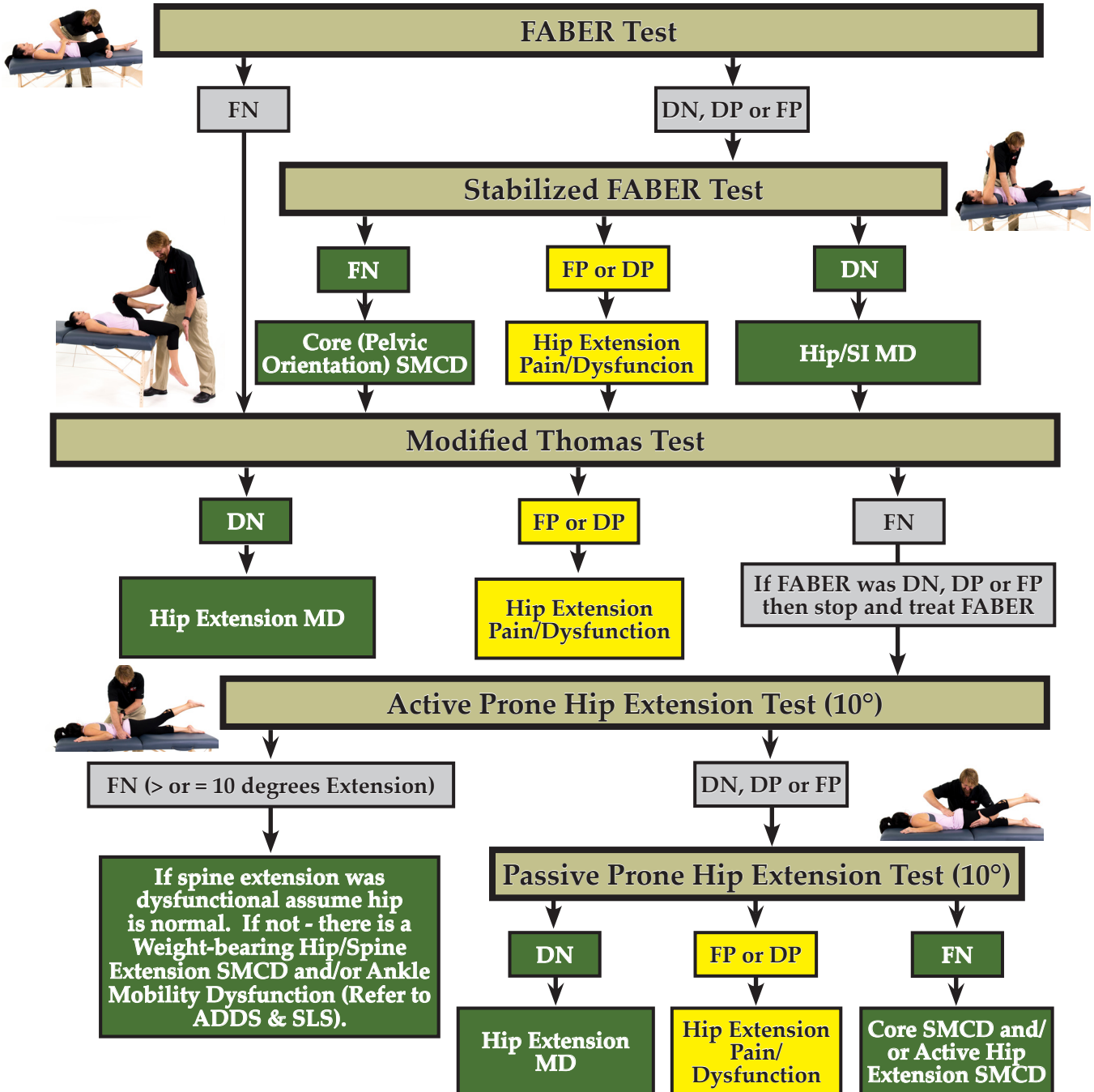
## Spine/Upper Body Flowchart



# MULTI-SEGMENTAL EXTENSION BREAKOUTS



## Lower Body Flowchart



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Prone Press Up Test:

To test spine extension in an unloaded posture.

### How to Perform the Prone Press Up Test:

Instruct the patient to lie prone with the head face down, arms by the sides with elbows bent, and thumbs in line with their clavicles. Have the person push to a straight arm position, extending the trunk up as high as possible.

### Functional Criteria:

Elbows should fully extend at the end of the movement with their ASIS remaining within 2 inches of the test surface. They should also demonstrate a uniform spinal curve.

### Additional Information:

It is acceptable for the ASIS to lift up to 2 inches off the ground to remain within normal limits. An Airex pad can be used to help measure. When at end range press up, you should not be able to slide the Airex pad under the patient's ASIS. If you can, they lifted more than two inches.

Pay attention to hand position as this can affect the results of this test.

### Possible Findings:

- FN: Ability to complete the movement with a uniform spine curve, no pain
  - » If the finding is FN - that is, the patient can perform full spine extension in the prone position but not while standing - the patient might have a weight-bearing spine extension stability/motor control dysfunction. But the patient might also have a shoulder flexion finding. Proceed to the Active Prone Shoulder Girdle Flexion Test.
- FP: Ability to complete the movement with a uniform spine curve, pain
  - » Proceed to the Active Lumbar Locked (IR) Extension/Rotation Test.
- DN: Inability to perform the movement, no pain
  - » Proceed to the Active Lumbar Locked (IR) Extension/Rotation Test.
- DP: Inability to perform the movement, pain
  - » Proceed to the Active Lumbar Locked (IR) Extension/Rotation Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Active Lumbar Locked (IR) Extension/Rotation Test:

To test the combined active extension and rotation in the thorax in an unloaded posture.

### How to Perform the Active Lumbar Locked (IR) Extension/Rotation Test:

Instruct the patient to get into a full prone rocking position (butt to heels) with their forearms together in front of their knees. Their knees do not need to be together. Place their right hand and forearm behind on the small of their back with their palm facing upwards. To set their scapula in proper position, the shoulder should be in a mid-axial position. Their left forearm and hand remain on the testing surface in the start position. Have the person rotate their head and thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

### Functional Criteria:

When using a line through the AC Joints as the reference line, normal thorax extension/rotation is 50 degrees relative to the horizontal bilaterally, without lateral flexion/side bending.

### Additional Information:

The prone rocking position helps eliminate the lumbar spine from contributing to the extension/rotation. This can also differentiate between right and left thorax restrictions.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

### Possible Findings:

- FN: 50 degrees or greater of active thorax extension/rotation, no pain
  - » Consider thorax extension/rotation normal. Proceed to the Active Prone on Elbow Extension/Rotation Test.
- FP: 50 degrees or greater of active thorax extension/rotation, pain
  - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation Test.
- DN: Less than 50 degrees of passive thorax extension/rotation on one side (unilateral DN), or on both sides (bilateral DN) no pain
  - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation Test.
- DP: Less than 50 degrees of passive thorax extension/rotation on one side (unilateral DP), or on both sides (bilateral DP), pain
  - » Proceed to Passive Lumbar Locked (IR) Extension/Rotation Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Passive Lumbar Locked (IR) Extension/Rotation Test:

To test the combined passive extension and rotation of the thorax in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

### How to Perform the Passive Lumbar Locked (IR) Extension/Rotation Test:

Instruct the patient to get into a full prone rocking position (butt to heels) with their forearms together in front of their knees. Place their right hand and forearm behind on the small of their back with their palm facing upwards. To set their scapula in proper position, the shoulder should be in a mid-axial position. Their left forearm and hand remain on the testing surface in the start position. Stabilize on the outside of their left shoulder with your left hand and at their hip, by placing your right hip against their left hip. This allows you to create a tunnel for rotation. Thread your right arm through their elbow, placing your right forearm tightly along their trunk with the ulnar border of your hand over their coracoid process. Passively rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

### Functional Criteria:

When using a line through the AC Joints as the reference line, normal thorax extension/rotation is 50 degrees relative to the horizontal bilaterally, without lateral flexion/side bending.

### Additional Information:

The prone rocking position helps eliminate the lumbar spine from contributing to the extension/rotation. This can also differentiate between right and left upper thorax restrictions.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting their right hand on their left shoulders.

### Possible Findings:

- FN: 50 degrees or greater of passive thorax extension/rotation, no pain
  - » Indicates a thorax extension/rotation stability/motor control dysfunction. Proceed to the Active Prone on Elbows (IR) Extension/Rotation Test.
- FP: 50 degrees or greater of passive thorax extension/rotation, pain
  - » Indicates pain with thorax extension/rotation - but this could be due to Lumbar mobility problem. Proceed to the Prone on Elbow Test.
- DN: Less than 50 degrees of passive thorax extension/rotation on one side (unilateral DN) or both sides (bilateral DN), no pain
  - » Indicates a unilateral/bilateral thorax extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Shoulder Girdle Flexion Test.
- DP: Less than 50 degrees of passive thorax extension/rotation on one side (unilateral DP) or both sides (bilateral DP), pain
  - » Indicates a painful unilateral/bilateral thorax extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Shoulder Girdle Flexion Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Active Prone on Elbow (IR) Extension/Rotation Test:

To test the active extension and rotation of the lumbar region in an unloaded posture.

### How to Perform the Active Prone on Elbow Unilateral Ext./Rot. (IR) Test:

Instruct the patient to lie prone, then push up to elbows and position forearms together. Place their right hand behind the back with their palms facing upward. The left forearm and hand remain in the start position. Have the person rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

### Functional Criteria:

When using a line through the AC Joints as the reference line, normal lumbar extension/rotation is 30 degrees relative to the floor bilaterally. The hips should remain in contact with the floor during movement.

### Additional Information:

Both the thorax and lumbar spine contribute to the 30 degrees of movement seen from the AC Joints - 20 degrees coming from the thorax and 10 degrees from the lumbar region.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting right hand on their left shoulder.

### Possible Findings:

- FN: 30 degrees or greater of extension/rotation, no pain
  - » If the finding is FN on both sides, there is still potential for a bilateral spine extension motor control dysfunction, but we must check shoulder flexion. Proceed to the Prone Shoulder Girdle Flexion Test.
- FP: 30 degrees or greater of extension/rotation, pain
  - » Proceed to the Passive Prone on Elbow Extension/Rotation Test.
- DN: Less than 30 degrees of extension/rotation on one side (unilateral DN) or on both sides (bilateral DN), no pain
  - » Proceed to the Passive Prone on Elbow Extension/Rotation Test.
- DP: Less than 30 degrees of extension/rotation on one side (unilateral DN) or on both sides (bilateral DN), pain
  - » Proceed to the Passive Prone on Elbow Extension/Rotation Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Passive Prone on Elbow (IR) Extension/Rotation Test:

To test the passive extension and rotation of the lumbar region in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

### How to Perform the Passive Prone on Elbow (IR) Extension/Rotation Test:

Instruct the patient to lie prone, then push up to elbows and position forearms together. Place their right hand behind the back with their palm facing upward. The left forearm and hand remain in the start position. Stabilize on the outside of their left shoulder with your left hand and at their hip, by placing your right hip against their left hip. This allows you to create a tunnel for rotation. Thread your right arm through their elbow, placing your right forearm tightly along their trunk with the ulnar border of your hand over their coracoid process. Passively rotate the thorax to the right as far as possible while maintaining the set-up position. Compare both sides.

### Functional Criteria:

When using a line through the AC Joints as the reference line, normal lumbar extension/rotation is 30 degrees relative to the floor bilaterally. The hips should remain in contact with the floor during movement.

### Additional Information:

Both the thorax and lumbar spine contribute to the 30 degrees of movement seen from the AC Joints - 20 degrees coming from the thorax and 10 degrees from the lumbar spine.

If the patient has pain with the shoulder IR test position, it is appropriate to use a chest position for the arm - their right arm across their chest, resting their right hand on their left shoulder.

### Possible Findings:

- FN: 30 degrees or greater of extension/rotation, no pain
  - » If the finding is FN on both sides, there is a lumbar extension/rotation stability/motor control dysfunction. Proceed to Active Prone Shoulder Girdle Flexion Test.
- FP: 30 degrees or greater of extension/rotation, pain
  - » Indicates pain with lumbar extension/rotation - complete local biomechanical exam. Proceed to the Active Prone Shoulder Girdle Flexion Test.
- DN: Less than 30 degrees of extension/rotation on one side (unilateral DP) or on both sides (bilateral DP), no pain
  - » Indicates a unilateral/bilateral lumbar extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Shoulder Girdle Flexion Test.
- DP: Less than 30 degrees of extension/rotation on one side (unilateral DP) or on both sides (bilateral DP), pain
  - » Indicates a painful unilateral/bilateral lumbar extension/rotation mobility dysfunction - complete local biomechanical exam. Proceed to the Active Prone Shoulder Girdle Flexion Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Active Prone Shoulder Flexion Test:

To test active shoulder flexion mobility and stability in an unloaded posture.

### How to Perform the Active Prone Shoulder Flexion Test:

Instruct the patient to assume the prone position with the arms and hands hanging down off the table. The patient then attempts to flex their arm up as high as possible into full shoulder flexion.

### Functional Criteria:

Normal range for shoulder flexion is 170 degrees.

### Additional Information:

Make sure the patient keeps their torso in contact with the table - no thorax extension or rotation.

### Possible Findings:

- FN: 170 degrees or greater of active shoulder girdle flexion, no pain
  - » Indicates a potential postural upper quarter stability/motor control dysfunction. Proceed to the Lower Body Extension Flowchart.
- FP: 170 degrees or greater of active shoulder girdle flexion, pain
  - » Proceed to the Passive Prone Shoulder Flexion Test.
- DN: Less than 170 degrees of active shoulder girdle flexion, no pain
  - » Proceed to the Passive Prone Shoulder Flexion Test.
- DP: Less than 170 degrees of active shoulder girdle flexion, pain
  - » Proceed to the Passive Prone Shoulder Flexion Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Passive Prone Shoulder Flexion Test:

To test passive shoulder flexion in an unloaded posture to differentiate between mobility and stability/motor control dysfunction.

### How to Perform the Passive Prone Shoulder Flexion Test:

Instruct the patient to assume the prone position with the arms and hands hanging down off the table. Passively flex their arm up as high as possible into full shoulder flexion.

### Functional Criteria:

Normal range for shoulder flexion is 170 degrees.

### Additional Information:

Make sure the patient keeps their torso in contact with the table - no thorax extension or rotation.

### Possible Findings:

- FN: 170 degrees or greater of passive shoulder girdle flexion, no pain
  - » Indicates a shoulder flexion stability/motor control dysfunction. Proceed to the Lower Body Extension Flowchart.
- FP: 170 degrees or greater of passive shoulder girdle flexion, pain
  - » Indicates pain with shoulder flexion - complete local biomechanical exam. Proceed to the Lower Body Extension Flowchart.
- DN: Less than 170 degrees of passive shoulder girdle flexion, no pain
  - » Indicates a shoulder flexion mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Body Extension Flowchart.
- DP: Less than 170 degrees of passive shoulder girdle flexion, pain
  - » Indicates a painful shoulder flexion/abduction mobility dysfunction - complete local biomechanical exam. Proceed to the Lower Body Extension Flowchart.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the FABER Test:

To test the effect of hip flexion, abduction and external rotation positioning on the hip and SI's ability to extend.

### How to Perform the FABER Test:

Instruct the patient to lie supine and then flex, abduct and externally rotate the left hip by placing the left foot over the right thigh or knee. Stabilize the pelvis on the right side. Have the patient slowly lower the left knee down toward the table. You can also apply passive overpressure. Look for restrictions or signs of pain. Repeat on the other side.

### Functional Criteria:

Normal movement is the patient's knee to be within two of their fist-widths from the table/floor without pain.

### Additional Information:

The name of this test tells us the set-up position—F-flexion, AB-duction, E-ternal R-otation of the hip. Once there, we extend the hip. If there's pain in the front of the hip, joint discomfort or limited motion, suspect hip joint pathology such as degenerative joint disease (DJD), anterior hip capsular or labrum tightness. If pain develops in the back over the lumbar spine or SI joint, suspect limited hip joint mobility creating lumbo-pelvic instability.

### Possible Findings:

- FN: Knee is within 2 of their fist-widths from the table/floor, no pain
  - » Proceed to the Modified Thomas Test.
- FP: Knee is within 2 of their fist-widths from the table/floor, pain
  - » Proceed to the Stabilized FABER Test.
- DN: Knee is greater than 2 of their fist-widths from the table/floor, no pain
  - » Proceed to the Stabilized FABER Test.
- DP: Knee is greater than 2 of their fist-widths from the table/floor, pain
  - » Proceed to the Stabilized FABER Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Stabilized FABER Test:

To test if there is an underlying SMCD of the core that may be limiting hip mobility. Pelvic orientation can directly affect the spacing in the acetabulum and therefore give the impression that there is a hip joint mobility problem, when in reality it may just be a stability/motor control problem.

### How to Perform the Stabilized FABER Test:

Instruct the patient to lie supine with their arms extended out in front of them towards the ceiling, and then flex, abduct and externally rotate the left hip by placing the left foot over the right thigh or knee. Stabilize the pelvis on the right side. Now, have the patient push against resistance with their hands towards their feet to activate the core and create a temporary posterior tilt of the pelvis. Next, have the patient slowly lower the knee down toward the table, essentially repeating the positive FABER Test. Look for any change in the restriction or amount of pain. Repeat on the other side.

### Functional Criteria:

Normal movement is the patient's knee to be within two of their fist-widths from the table/floor without pain.

### Additional Information:

You can activate the core by resisting manually with your hands against the patient's hands or use some elastic tubing to create the resistance.

### Possible Findings:

- FN: Knee is within 2 of their fist-widths from the table/floor, no pain
  - » Positive for Hip/Core stability/motor control dysfunction. Proceed to the Modified Thomas Test.
- FP: Knee is within 2 of their fist-widths from the table/floor, pain
  - » Indicates pain in the hip/SI region - complete local biomechanical exam. Proceed to Modified Thomas Test.
- DN: Knee is greater than 2 of their fist-widths from the table/floor, no pain
  - » Indicates a Hip/SI mobility dysfunction - complete local biomechanical exam. Proceed to Modified Thomas Test.
- DP: Knee is greater than 2 of their fist-widths from the table/floor, pain
  - » Indicates a painful Hip/SI mobility dysfunction - complete local biomechanical exam. Proceed to Modified Thomas Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Modified Thomas Test:

To test the overall mobility of the anterior and lateral tissues of the hip and the anterior hip capsule.

### How to Perform the Modified Thomas Test

Have the patient sit on the edge of a treatment table and roll backward, bringing knees to chest while rolling back. Find neutral pelvic position by flexing the hips, being careful to not posterior tilt the pelvis by taking knees too far into their chest. Either hold one knee in place or have patient assist by holding their own knee in position. Make sure neutral pelvis is maintained.

Take the non-supported leg and place it in 90 degrees of hip flexion and 90 degrees of knee flexion, and adduct it next to the supported leg. Passively lower the leg as far as possible.

### Functional Criteria:

A normal Thomas test is thigh parallel to or below the table with the knee maintaining 90 degrees without abduction at the hip joint.

### Additional Information:

Ask the patient to place zero tension in the leg that's being lowered. Make sure the patient is not arching the back during the lowering phase and keeps a firm grasp of the opposite leg.

### Possible Findings:

- FN: Thigh is parallel to the table or below, with 90 degrees of knee flexion and no hip abduction, no pain
  - » If FABER Test was DN, DP, or FP, then stop and treat FABER. If FABER Test was FN, proceed to Active Prone Hip Extension Test.
- FP: Thigh is parallel to the table or below, with 90 degrees of knee flexion and no hip abduction, pain
  - » Indicates pain with hip extension - complete local biomechanical exam.
- DN: Thigh unable to reach parallel to the table or below and/or less than 90 degrees of knee flexion and/or hip abduction, no pain
  - » Indicates a hip extension mobility dysfunction - complete local biomechanical exam.
- DP: Thigh unable to reach parallel to the table or below and/or less than 90 degrees of knee flexion and/or hip abduction, pain
  - » Indicates a painful hip extension mobility dysfunction - complete local biomechanical exam.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Active Prone Hip Extension Test:

To test active hip extension beyond 0 degrees of extension.

### How to Perform the Active Prone Hip Extension Test:

Instruct the patient to lie prone with the hands down by the sides or under the head. Have the patient actively extend the hip.

### Functional Criteria:

Normal movement is 10 degrees of hip extension without lumbar extension

### Additional Information:

Watch for anterior tilt of the pelvis - palpating the ASIS can help identify if anterior tilting is occurring prior to 10 degrees. Also watch for external rotation and abduction of the hip.

### Possible Findings:

- FN: 10 degrees of active hip extension, no pain
  - » If spine extension was dysfunctional, consider the hip normal. If not, indicates a weight-bearing hip extension stability/motor control dysfunction and/or limited ankle dorsiflexion.
- FP: 10 degrees of active hip extension, pain
  - » Proceed to the Passive Prone Hip Extension Test.
- DN: Less than 10 degrees of active hip extension, no pain
  - » Proceed to the Passive Prone Hip Extension Test.
- DP: Less than 10 degrees of active hip extension, pain
  - » Proceed to the Passive Prone Hip Extension Test.



## MULTI-SEGMENTAL EXTENSION BREAKOUTS



### Test Objective for the Passive Prone Hip Extension Test:

To test passive hip extension beyond 0 degrees of extension to differentiate between mobility and stability/motor control dysfunction.

### How to Perform the Passive Prone Hip Extension Test:

Instruct the patient to lie prone with the hands down by the sides or under the head. Passively extend the right hip as far as possible.

### Functional Criteria:

Normal movement is 10 degrees or greater of hip extension without lumbar extension.

### Additional Information:

Passively we are looking for 10 degrees or greater of hip extension without anterior tilt of the pelvis or external rotation and abduction of the foot.

### Possible Findings:

- FN: 10 degrees or greater of passive hip extension, no pain
  - » Indicates a core and/or an active hip extension stability/motor control dysfunction.
- FP: 10 degrees or greater of passive hip extension, pain
  - » Indicates pain with hip extension - complete local biomechanical exam.
- DN: less than 10 degrees of passive hip extension, no pain
  - » Indicates a hip extension mobility dysfunction - complete local biomechanical exam.
- DP: less than 10 degrees of passive hip extension, pain
  - » Indicates a painful hip extension mobility dysfunction - complete local biomechanical exam.

